CE 310 — Week 8 Midterm Study Guide

Covers: Weeks 1–7
Exam format: Session 2 (Wed Oct 14) = Concept Exam (closed computer, ~50 min, 50 pts); Session 3 (Fri Oct 16) = In-Class Exercise Practicum (open Excel, open notes, ~50 min, 100 pts)

Week 1 — The Flaw of Averages and TMY3 Data

Core concept. Using the average value of a variable to represent the whole distribution leads to systematic errors whenever the relationship between input and output is nonlinear. For equipment sizing, the peak (extreme) matters more than the mean.

Key Excel functions

Function	Syntax	What it does
AVERAGE	`=AVERAGE(range)`	Arithmetic mean of all values in range
MAX	`=MAX(range)`	Largest value in range
MIN	`=MIN(range)`	Smallest value in range

Worked example. Tucson outdoor temperature has μ = 68.75°F and MAX ≈ 106.9°F. A chiller sized for 68.75°F would fail on any summer day above its rated capacity. The MAX (or a high design percentile) drives sizing decisions, not the mean.

Common mistake. Reporting only the mean and concluding “the system is adequate on average.” Average-adequate does not mean peak-adequate. Always check the tail.

Week 2 — Descriptive Statistics: Spread and Shape

Core concept. Standard deviation, skewness, and IQR describe how spread out and asymmetric a distribution is. Histograms and box plots reveal shape that summary statistics alone can hide.

Key Excel functions

Function	Syntax	What it does
STDEV	`=STDEV(range)`	Sample standard deviation (ddof=1)
SKEW	`=SKEW(range)`	Fisher skewness — positive = right tail

Function	Syntax	What it does
QUARTILE.INC	`=QUARTILE.INC(range, 1)`	First quartile (Q1)
QUARTILE.INC	`=QUARTILE.INC(range, 3)`	Third quartile (Q3)

IQR = Q3 − Q1. Compute in Excel as `=QUARTILE.INC(range,3) − QUARTILE.INC(range,1)`.

Worked example. Cooling_kWh in the Tucson dataset has σ = 25.3 kWh and positive skew (long right tail — a few peak cooling hours are much larger than the median). The IQR is tighter than 1.35 × σ would predict, confirming the right skew.

Common mistake. Confusing STDEV (sample, ddof=1) with STDEVP (population, ddof=0). In engineering data where you have a sample, always use STDEV.

Week 3 — Conditional Probability and Pivot Tables

Core concept. Conditional probability P(A | B) answers "given that B occurred, what is the probability A also occurred?" It is computed as the joint count divided by the conditioning event count.

Key Excel functions

Function	Syntax	What it does
COUNTIF	`=COUNTIF(range, criteria)`	Count cells meeting one condition
COUNTIFS	`=COUNTIFS(range1, crit1, range2, crit2)`	Count cells meeting multiple conditions

P(A | B) = COUNTIFS(A_range, A_crit, B_range, B_crit) / COUNTIF(B_range, B_crit)

Worked example. In the Tucson dataset: P(BusinessHours = "Yes" | Season = "Summer") = COUNTIFS(Season, "Summer", BusinessHours, "Yes") / COUNTIF(Season, "Summer"). Pivot tables automate this for all combinations simultaneously.

Common mistake. Dividing by the total (8,760) instead of by the conditioning count. P(A | B) denominates on B, not on the whole dataset.

Reversing the conditional — Bayes. P(A | B) and P(B | A) are *different numbers*, because they divide the same joint count by different denominators. Reversing one and quoting it as the other is the most common error in this topic.

P(A | B) = P(B | A) × P(A) / P(B), where the denominator sums every path to B:
P(B) = P(B | A) × P(A) + P(B | not A) × P(not A)

Worked example. A fault detector fires on 90% of faulty units and on 4% of healthy ones. If 5% of units are faulty, then given an alarm: P(Fault | Alarm) = (0.90 × 0.05) / (0.90 × 0.05 + 0.04 × 0.95) = 0.045 / 0.083 = **54%** — not 90%. The alarm raises the probability from 5% to 54%, but because healthy units vastly outnumber faulty ones, nearly half of all alarms are still false.

Be ready to: compute a reversed conditional from counts or rates, and explain why a low base rate keeps the posterior far below the detector's hit rate.

Week 4 — Poisson and Binomial Distributions

Core concept. The Poisson distribution models the number of discrete events occurring in a fixed interval of time or space when the average rate λ is known and events are independent.

Key Excel functions

Function	Syntax	What it does
POISSON.DIST	`=POISSON.DIST(k, lambda, FALSE)`	P(X = k) — exact count (PMF)
POISSON.DIST	`=POISSON.DIST(k, lambda, TRUE)`	P(X ≤ k) — cumulative (CDF)

Worked example. If λ = 3 equipment failures per month, then P(X = 0) = POISSON.DIST(0, 3, FALSE) = e^−3 ≈ 0.0498. There is about a 5% chance of zero failures in a given month.

Common mistake. Using λ for the wrong time window. If the rate is 6 per hour and you want 15-minute probability, use λ = 6/4 = 1.5 per 15-minute interval.

The binomial — a fixed number of trials. Use the binomial when you have *n* independent trials, each succeeding with the same probability *p*, and you want exactly *k* successes:

P(X = k) = C(n, k) × p^k × (1 − p)^(n − k), where C(n, k) = n! / [k! (n − k)!]

In Excel: `=BINOM.DIST(k, n, p, FALSE)` for exactly k, `TRUE` for k or fewer.

Worked example. 8 valves, each with an independent 15% chance of failing inspection. P(exactly 2 fail) = C(8,2) × 0.15^2 × 0.85^6 = 28 × 0.0225 × 0.3771 = **0.238 (23.8%)**.

Choosing between them. Binomial needs a fixed *n* and a per-trial *p* ("5 sensors, 10% each"). Poisson needs only a rate over an interval, with no fixed number of trials ("0.5 overflows per year"). Poisson is the limit of the binomial when *n* is large and *p* is small.

Common mistake. Forgetting the C(n, k) term. Without it you compute the probability of one *specific* arrangement, not of any k of the n — which undercounts by a factor of C(n, k).

Be ready to: write a binomial expression by hand, evaluate it, and say why the independence assumption might fail on a real site (shared batch, shared power supply, same installer, same environment).

Week 5 — Normal and Lognormal Distributions

Core concept. The Normal distribution (bell curve) is symmetric. The Lognormal distribution models variables that are always positive with a right skew (e.g., construction cost per cubic yard — use AVERAGE(LN(range)) and STDEV(LN(range)) for the log-space parameters). Z-scores standardize any Normal variable: $z = (x - \mu) / \sigma$.

Key Excel functions

Function	Syntax	What it does
NORM.DIST	`=NORM.DIST(x, mean, std, TRUE)`	P(X ≤ x) — left-tail area (CDF)
NORM.DIST	`=NORM.DIST(x, mean, std, FALSE)`	Probability density at x (PDF)
NORM.INV	`=NORM.INV(p, mean, std)`	x such that P(X ≤ x) = p
LOGNORM.DIST	`=LOGNORM.DIST(x, ln_mean, ln_std, TRUE)`	Lognormal CDF

68-95-99.7 rule: 68% of observations fall within ±1σ, 95% within ±2σ, 99.7% within ±3σ.

Worked example. Beta supplier concrete: μ = 4,367 psi, σ = 484 psi. P(fc < 4,000) = NORM.DIST(4000, 4367, 484, TRUE) = 0.224 = 22.4%. Beta fails ACI 318 (criterion: no more than 10% of cylinders below 4,000 psi). Alpha: μ

= 4,632 psi, σ = 344 psi → NORM.DIST(4000, 4632, 344, TRUE) ≈ 3.3% — compliant.

Common mistake. Using NORM.DIST with `FALSE` (density) when you want a probability. Only the `TRUE` (cumulative) form returns a probability.

Week 6 — Empirical Percentiles and Exceedance Probability

Core concept. Empirical percentiles are computed directly from ranked data using LARGE or PERCENTILE. Exceedance probability is P(X > threshold) = 1 − P(X ≤ threshold), often expressed as a fraction of the dataset above a threshold.

Key Excel functions

Function	Syntax	What it does
LARGE	`=LARGE(range, k)`	k-th largest value
SMALL	`=SMALL(range, k)`	k-th smallest value
PERCENTILE	`=PERCENTILE(range, p)`	Value at the p-th percentile (0 ≤ p ≤ 1)
COUNTIF	`=COUNTIF(range, ">threshold")`	Count values above a threshold

LARGE(data, k) returns the k-th largest value. In a 500-row dataset: LARGE(data, 5) = empirical 99th %ile (5 = 1% × 500); LARGE(data, 75) = empirical 85th %ile (75 = 15% × 500).

Worked example. Traffic speed (500 rows): NORM.INV(0.85, 54.4, 7.8) = 62.5 mph (Normal model 85th %ile). Empirical equivalent: LARGE(Speed_mph, 75) (15% × 500 = 75 rows). The two methods agree near the 85th percentile; divergence grows toward the extreme tails where the Normal assumption is least reliable.

Common mistake. Using PERCENTILE(data, 0.85) and LARGE(data, 75) expecting identical results in a 500-row dataset. PERCENTILE interpolates; LARGE returns an actual data value. They will be close but not identical.

Week 7 — Correlation and Covariance

Core concept. Pearson r measures the strength and direction of a linear relationship between two variables (−1 ≤ r ≤ 1). r^2 is the coefficient of determination: the fraction of variance in Y explained by X. Covariance has units; r is dimensionless. r = cov(X, Y) / (σ_X × σ_Y).

Key Excel functions

Function	Syntax	What it does
CORREL	`=CORREL(range1, range2)`	Pearson r between two arrays
COVARIANCE.S	`=COVARIANCE.S(range1, range2)`	Sample covariance (ddof=1)
FILTER	`=FILTER(range, condition)`	Extract rows matching a condition

Conditional CORREL: `=CORREL(FILTER(range1, condition), FILTER(range2, condition))`

Worked example. r(Outdoor_Temp_F, Cooling_kWh) = 0.7752 for all 8,760 Tucson hours. During business hours only: r = 0.9539. The business-hours r is stronger because the HVAC system runs at full capacity with a predictable outdoor-cooling relationship. $r^2 = 0.60$ all-hours → outdoor temperature explains 60% of cooling energy variance.

Common mistake. Interpreting r = 0 as "no relationship." r = 0 only means no LINEAR relationship. A strong U-shaped or other nonlinear relationship can produce $r \approx 0$.

Formula Sheet

Function	Syntax	What it returns	Week introduced
AVERAGE	`=AVERAGE(range)`	Arithmetic mean	W1
MAX	`=MAX(range)`	Largest value	W1
MIN	`=MIN(range)`	Smallest value	W1
STDEV	`=STDEV(range)`	Sample standard deviation	W2
SKEW	`=SKEW(range)`	Fisher skewness coefficient	W2
QUARTILE.INC	`=QUARTILE.INC(range, quartile_num)`	Q1 (1) or Q3 (3)	W2
COUNTIF	`=COUNTIF(range, criteria)`	Count meeting one condition	W3
COUNTIFS	`=COUNTIFS(range1, crit1, range2, crit2)`	Count meeting multiple conditions	W3

Function	Syntax	What it returns	Week introduced
POISSON.DIST	`=POISSON.DIST(k, lambda, FALSE)`	P(X = k) exact Poisson	W4
POISSON.DIST	`=POISSON.DIST(k, lambda, TRUE)`	P(X ≤ k) cumulative Poisson	W4
NORM.DIST	`=NORM.DIST(x, mean, std, TRUE)`	P(X ≤ x) normal CDF	W5
NORM.DIST	`=NORM.DIST(x, mean, std, FALSE)`	Normal probability density	W5
NORM.INV	`=NORM.INV(p, mean, std)`	Inverse normal (find x given p)	W5
LOGNORM.DIST	`=LOGNORM.DIST(x, ln_mean, ln_std, TRUE)`	Lognormal CDF	W5
LARGE	`=LARGE(range, k)`	k-th largest value	W6
SMALL	`=SMALL(range, k)`	k-th smallest value	W6
PERCENTILE	`=PERCENTILE(range, p)`	Value at p-th percentile	W6
CORREL	`=CORREL(range1, range2)`	Pearson r (dimensionless)	W7
COVARIANCE.S	`=COVARIANCE.S(range1, range2)`	Sample covariance (with units)	W7
FILTER	`=FILTER(range, condition)`	Subset of range matching condition	W7
AVERAGEIF	`=AVERAGEIF(crit_range, criteria, avg_range)`	Conditional mean	W3/W7